Non-Uniform Accn Worksheet

Class 11 Physics | Nine Education IIT Academy | Kinematics — Variable Acceleration

Name: ____________________ Date: ____________

The Three Working Relations

1	**$v = dx/dt$**	Links position and velocity directly — use to move between $x(t)$ and $v(t)$.
2	**$a = dv/dt$**	Use when acceleration is a function of **time**: $a = f(t)$. Integrate over t.
3	**$a = v\cdot(dv/dx)$**	Use when acceleration is a function of **position**: $a = f(x)$. No t needed — separate v dv from a(x) dx and integrate with limits.

Take all quantities in SI units. Where a numeric answer is irrational, leave it in surd form or round to 2 decimal places.

Formula 1: v = dx/dt — Position ↔ Velocity

1. The position of a particle is $x = 4t^2 - 3t + 2$ (m). Find its velocity and acceleration as functions of time.

2. A particle moves with $v(t) = 6t^2 - 4t$ (m/s). If $x = 3$ m at $t = 0$, find its position at $t = 2$ s.

3. The position of a particle is $x = t^3 - 6t^2 + 9t$ (m). Find the time(s) at which the particle is momentarily at rest.

4. The position of a particle is $x = 5t^2 - 2t^3$ (m). Find the velocity at $t = 1$ s, and the time ($t > 0$) at which the velocity becomes zero.

5. A particle's velocity is $v = 4t + 1$ (m/s). If $x = 2$ m at $t = 1$ s, find its position at $t = 3$ s.

6. The displacement of a particle is $x = 2t^3 - 5t + 1$ (m). Find its acceleration at $t = 2$ s.

7. The position of a particle is $x = t^2 - 4t + 4$ (m). Find the time at which the particle is at the origin, and its velocity at that instant.

Formula 2: a = dv/dt — When a = f(t)

8. A particle starts from rest at the origin with acceleration $a = 4t$ (SI). Find its velocity and position at $t = 3$ s.

9. A particle has acceleration $a = (6 - 2t)$ m/s² and starts with $u = 0$ at $t = 0$ from the origin. Find the time at which its velocity is maximum, and that maximum velocity.

10. A particle starts with $u = 5$ m/s and acceleration $a = 3t^2$ (SI). Find its velocity at $t = 2$ s and its displacement in that time (starting at $x = 0$).

11. A particle experiences acceleration $a = 2 + 6t$ (SI), starting from rest at $x = 2$ m. Find the velocity and position at $t = 2$ s.

12. The acceleration of a particle is $a = (6t - 6)$ m/s², and $v = 0$ at $t = 0$. Find $v(t)$, the other time ($t > 0$) at which the particle is momentarily at rest, and its velocity at $t = 1$ s.

13. A particle moves with acceleration $a = (4 - t^2)$ m/s², starting from rest at the origin. Find its velocity and position at $t = 2$ s.

14. A particle's acceleration is $a = (2t + 3)$ m/s². It starts with $u = 2$ m/s at $t = 0$ from $x = 0$. Find the velocity at $t = 3$ s and the distance covered in that time.

15. The acceleration of a particle varies as $a = (3t^2 - 4t)$ m/s². If it starts from rest at the origin, find its displacement between $t = 1$ s and $t = 3$ s.

16. A rocket accelerates as $a = (20 - 4t)$ m/s² from rest at ground level. Find the time at which its velocity is maximum, the maximum velocity, and the height reached at that time.

Formula 3: $a = v(dv/dx)$ — When $a = f(x)$

17. A particle has acceleration $a = 9x$ (SI). Starting from rest at $x = 0$, find its speed at $x = 2$ m.

18. A particle has acceleration $a = -8x$ (SI), with $v = 0$ at $x = 3$ m. Find its speed at $x = 1$ m.

19. A particle experiences acceleration $a = 6x + 4$ (SI) and has speed 3 m/s at $x = 0$. Find its speed at $x = 2$ m.

20. A particle has acceleration $a = -2x$ (SI). At $x = 0$, its speed is 4 m/s. Find its speed at $x = 1$ m.

21. A particle starts from rest at the origin with acceleration $a = 6x^2$ (SI). Find its speed when it reaches $x = 1$ m.

22. A block released from rest at $x = 0$ experiences acceleration $a = 2x$ (SI). Find its speed when it reaches $x = 4$ m.

23. A particle's velocity is related to its position by $v^2 = 16 - 4x^2$ (SI). Find its acceleration as a function of x.

24. A particle is momentarily at rest at $x = 0$ and has acceleration $a = (8 - 2x)$ m/s². Find its speed at $x = 2$ m.

25. The velocity of a particle varies with position as $v = 2x$ (SI). Find the acceleration as a function of x, and its value at $x = 3$ m.

Hard / Challenge

26. A particle moves with acceleration $a = -kv$ (resistive), starting with velocity v_0 at $t = 0$. Show that $v = v_0e^{-kt}$, and find the time at which the velocity drops to half of v_0, in terms of k.

27. A particle's acceleration is $a = 2x + 3$ (SI). Starting from rest at $x = 0$, find its speed at $x = 2$ m.

28. The velocity of a particle varies with position as $v = 3x^2$ (SI). Find the acceleration as a function of x, and evaluate it at $x = 2$ m.

29. A particle's position is $x = t^3 - 6t^2 + 9t$ (m) — the same function as Q3. Find the total distance travelled between $t = 0$ s and $t = 4$ s (the particle reverses direction during this interval).

30. A particle's acceleration varies with position as $a = (-4x + 8)$ m/s², and it is momentarily at rest at $x = 1$ m. Show that it is also momentarily at rest at $x = 3$ m, and find its maximum speed and the position at which it occurs.

Non-Uniform Accn Worksheet — Answer Key

Formula 1 (Q1–7)

1. $v = (8t - 3)$ m/s; $a = 8$ m/s² (constant)

2. $x(2) = 11$ m

3. $t = 1$ s and $t = 3$ s

4. $v(1) = 4$ m/s; velocity = 0 at $t = 5/3$ s

5. $x(3) = 20$ m

6. $a(2) = 24$ m/s²

7. $t = 2$ s (a double root — the particle just touches the origin); $v(2) = 0$

Formula 2 (Q8–16)

8. $v(3) = 18$ m/s, $x(3) = 18$ m

9. $t = 3$ s, $v_{max} = 9$ m/s

10. $v(2) = 13$ m/s, $x(2) = 14$ m

11. $v(2) = 16$ m/s, $x(2) = 14$ m

12. $v(t) = 3t^2 - 6t$; rest again at $t = 2$ s; $v(1) = -3$ m/s

13. $v(2) = 16/3 \approx 5.33$ m/s; $x(2) = 20/3 \approx 6.67$ m

14. $v(3) = 20$ m/s; $x(3) = 28.5$ m

15. $\Delta x = 8/3 \approx 2.67$ m

16. $t = 5$ s, $v_{max} = 50$ m/s, height = $500/3 \approx 166.7$ m

Formula 3 (Q17–25)

17. $v = 6$ m/s

18. $v = 8$ m/s

19. $v = 7$ m/s

20. $v = \sqrt{14} \approx 3.74$ m/s

21. $v = 2$ m/s

22. $v = 4$ m/s

23. $a = -4x$ m/s²

24. $v = 2\sqrt{6} \approx 4.90$ m/s

25. $a = 4x$; $a(3) = 12$ m/s²

Hard / Challenge (Q26–30)

26. $t = (\ln 2)/k$

27. $v = 2\sqrt{5} \approx 4.47$ m/s

28. $a = 18x^3$; $a(2) = 144$ m/s²

29. total distance = 12 m (net displacement = 4 m)

30. rest confirmed at $x = 3$ m; maximum speed = 2 m/s, occurring at $x = 2$ m